RW–920

Sub. Code
511101

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Mathematics

ALGEBRA — I

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. If every element of the group G is its own inverse, prove that G is abelian.

2. If $a \in G$ and $a^m = e$, prove that $O(a) \mid m$.

3. Define inner automorphism of a group. Give an example.

4. List all the conjugate classes in S_3.

5. Describe all finite abelian groups of order 2^6.

6. If G is finite, prove that $a \in Z$ if and only if $O(N(a)) = O(G)$.

7. Let R be a ring. Prove that $a(-b) = -ab$ and $(-1)a = -a$ $\forall a, b \in R$.

8. Define division ring. Give an example.

9. Show that a Euclidean ring possesses a unit element.

10. State Gauss lemma.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Let H be a finite subset of a group G. Show that H is a subgroup if and only if H is a subgroup.

Or

(b) Show that a subgroup N of G is normal in G if and only if every left coset of N in G is a right coset of N in G.

12. (a) Define kernel of a group homomorphism and prove that it is a normal subgroup.

Or

(b) If G is a group, the set of all automorphisms of G is also a group.

13. (a) Show that a group of order 72 is not simple.

Or

(b) State and prove second part of Sylow's theorem.

14. (a) Let R be a commutative ring with unit whose only ideals are (O) and R itself. Prove that R is a field.

Or

(b) Prove that an ideal I of $\mathbf{Z}$ is prime if and only if $I = (p)$ where p is a prime number.

15. (a) Show that the gcd of any two elements in a Euclidean ring exists.

Or

(b) If p is a prime number of the form $4n+1$, prove that one can solve $x^2 \equiv -1 \mod p$.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries equal marks.

16. State and prove Lagrange's theorem.

17. Let G be a finite group. Prove that $O(G) = O(Z) + \sum_{a \notin Z} \frac{O(G)}{O(N(a))}$, where Z is the center of G.

18. Show that internal direct product is isomorphic to external direct product.

19. Show that every finite integral domain is a field.

20. If R is a unique factorization domain, show that $R[x]$ is also a unique factorization domain.

RW–921

Sub. Code
511102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Mathematics

REAL ANALYSIS

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give an example to show that arbitrary intersection of open sets need not be open.

2. Define compact set.

3. If $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = x^2$, show that f is continuous.

4. If $f:[a,b] \to \mathbb{R}$ is one-one and continuous, show that there exist $c, d \in \mathbb{R}$ such that $f([a,b]) = [c,d]$.

5. If $f(x)$ is defined for all real x and if for all real $x, y \left| f(x) - f(y) \right| \le (x-y)^2$, show that f is a constant.

6. Show that the function $f:[0, 2\pi] \to \mathbb{R}^2$ defined by $f(t) = (\cos t, \sin t)$ does not satisfy Mean-Value theorem.

7. State the necessary and sufficient condition for a bounded function to be Riemann stieltjes integrable.

8. Prove that $\int_a^b d\alpha = \alpha(b) - \alpha(a)$.

9. Show that $\sum \frac{x}{1+n^2x}$ converges uniformly on R.

10. If $f_n(x) = x^n$, $0 \le x \le 1$, show that f_n does not converge uniformly on $[0,1]$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that compact subset of a metric space set is closed.

Or

(b) Show that every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

12. (a) Show that continuous image of a compact set is compact.

Or

(b) If f is a continuous mapping of a metric space X into a metric space Y prove that $f(\overline{E}) \subset \overline{f(E)}$.

13. (a) If f is a real differentiable function on $[a,b]$ and if $f'(a) < \lambda < f'(b)$, show that there exists $x \in (a,b)$ such that $f'(x) = \lambda$.

Or

(b) State and prove Chain rule.

14. (a) If α increases monotonically on $[a,b]$ and if P^* is the refinement of the partition P of $[a,b]$ show that $U(P,f,\alpha) \geq U(P^*,f,\alpha)$.

Or

(b) If $f \in R$ on $[a,b]$, $x \in [a,b]$ then show that the function F defined by $F(x) = \int_a^x f(x)\,dx$ is differentiable whenever f is differentiable.

15. (a) If $f_n \to f$ uniformly and if each f_n is continuous at a show that f is continuous at a.

Or

(b) State and prove Weierstrass M-test.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that a subset E of $\mathbb{R}$ is connected iff for all $x, y \in E$ and $x < z < y$, then $z \in E$.

17. If f is a continuous mapping of a compact metric space X into a metric space Y, show that f is uniformly continuous.

18. State and prove L' Hospital's rule.

19. If $f \in R(\alpha)$ on $[a,b]$, $m \leq f \leq M, \phi$ continuous on $[m,M]$ and $h(x) = \phi(f(x))$ show that $h \in R(\alpha)$ on $[a,b]$.

20. If each f_n is real-valued differential function in (a,b), if there is at least one $x_0 \in (a,b)$ the sequence $\{f_n(x_0)\}$ converges and if $\{f_n'\}$ converges uniformly on (a,b), show that $\{f_n\}$ converges uniformly to a function f and $f'(x) = lin_{n\to\infty} f_n'(x)$.

RW–922

Sub. Code
511103

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Mathematics

ORDINARY DIFFERENTIAL EQUATIONS

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define linearly dependent on an interval I.

2. State Uniqueness theorem.

3. Verify that the function $\phi_1(x) = x(x > 0)$ satisfies the equation $x^2y''-xy'+y = 0$.

4. Write the Chebyshev equation.

5. Define singular point.

6. Write a short note on Bessel function of zero order of the first kind.

7. Define exact equation.

8. Write the Lipschitz condition.

9. State local existence theorem.

10. State the approximation and uniqueness theorem.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Let ϕ_1, ϕ_2 be any two linearly independent solutions of $L(y) = 0$ on I. Prove that every solution ϕ of $L(y) = 0$ can be written as $\phi = c_1\phi_1 + c_2\phi_2$, where c_1, c_2 are constants.

Or

(b) Compute the Wronskian of four linearly independent solutions of the equation $y^4 + 16y = 0$.

12. (a) Find two linearly independent power series solutions of the equation $y'' + y = 0$.

Or

(b) Show that $P_n(-x) = (-1)^n P_n(x)$ and hence that $P_n(-1) = (-1)^n$.

13. (a) Find the singular points of the following equations and determine these which are regular singular points (i) $x^2y'' - 5y' + 3x^2y = 0$ and (ii) $xy'' - 4y = 0$.

Or

(b) Prove that the series defining J_0 and K_0 converge for $|x| < \infty$.

14. (a) Find all real – valued solutions of the equation $y' = x^2y$.

Or

(b) Determine which equation are exact of the following:

(i) $2xydx + (x^2 + 3y^2)\,dy = 0$

(ii) $(x^2 + xy)dx + xy\,dy = 0$.

15. (a) State and prove uniqueness theorem.

Or

(b) Consider the initial value problem

$y_1^1 = y_2^2 + 1;\ y_2^1 = y_1^2$

$y_1(0) = 0,\ y_2(0) = 0$

Compute the first three successive approximations ϕ_0, ϕ_1, ϕ_2.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Show that $\phi_n(x) = \sin nx$ satisfies the boundary value problem $y''+n^2 y = 0,\ y(0) = 0,\ y(\pi) = 0$. where $n = 1, 2, 3, \ldots$.

(b) Using (a) show that $\int_0^{\pi} \sin nx \sin mx dx = 0$ if $n \neq m$.

17. Let ϕ_1 be a solution of $L(y) = 0$ on an internal I, and suppose $\phi_1(x) \neq 0$ on I. If $v_1, v_2, \ldots v_n$ is any basis on I for the solutions of the linear equations. $\phi_1 v^{n-1} + \cdots + \left[n\phi_1^{n-1} + (n-1)\phi_1^{n-1} + \cdots + a_{n-1}\phi_1\right]v = 0$ of value $(n-1)$ and if $v_k = v'_k$, $(k = 2, 3, \ldots n)$, then prove that $\phi_1, u_2\phi_1, \ldots, u_n\phi_1$ is a basis for the solution of $L(y) = 0$ on I.

18. Show that (a) $J_0'(x) = -J_1(x)$ (b) $K_0'(x) = -K_1(x)$.

19. Prove that a function ϕ is a solution of the initial value of problem $y' = f(x, y), y(x_0) = y_0$ on an interval I iff it is a solution of the integral equation $y = y_0 + \int_{y_0}^{x} f(t, y)\, dt$ on I.

20. Let f be a real – valued continuous function on the scrip $\delta : |x - x_0| \leqq a, \quad |y| < \infty, (a > 0)$, and suppose that f satisfies on δ a Lipschitz condition with constant $K > 0$. Prove that the successive approximations $\{\phi_K\}$ for the problem $y' = f(x, y),\ y(x_0) = y_0(x)$ exist on the infire internal $|x - x_0| \leqq a$ and a converge there to a solution ϕ of $(*)$.

RW–923

Sub. Code
511104

M.Sc. (Mathematics) DEGREE EXAMINATION, NOVEMBER 2017

First Semester

ANALYTICAL NUMBER THEORY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each questions carries 2 marks

1. State any four properties of divisibility.
2. If $(a,b) = (a,c) = 1$, prove that $(a,bc) = 1$.
3. State the Legendre's Identity.
4. If $x \geq 1$ prove that $\sum_{n>x} \frac{1}{n^3} = 0(x^{1-s})$, for $s < 1$.
5. Define the Chebyshev's functions $\psi(x)$ and $\upsilon(x)$.
6. State the Selberg's asymptotic formula.
7. Prove that congruence is an equivalence relation.
8. Solve the congruence $5x = 3 \pmod{24}$.
9. Define the Legendre's symbol.
10. State tile Quadratic reciprocity law.

Part B $(5 \times 5 = 25)$

Answer **all** questions by choosing either (a) or (b).

11. (a) Prove that for any two integers a and b there is a common divisor d of a and b of tile form $d = ax + by$.

Or

(b) If f and g are arithmetical functions prove the following :

(i) $(f+g)' = f' + g'$

(ii) $(f * g)' = f' * g + f * g'$

(iii) $(f^{-1})' = -f' * (f * f)^{-1}$, provided that $f(1) \neq 0$.

12. (a) State and prove Euler summation formula.

Or

(b) Prove that two lattice points (a,b) and (m,n) are mutually visible if, anti only if, $a-m$ and $b-n$ are relatively prime.

13. (a) State and prove t he Abel's identity.

Or

(b) Prove that for $n \geq 1$, the nth prime P_n satisfied the inequalities $\frac{1}{6} n \log n < P_n < 12\left(n \log n + n \log \frac{12}{e}\right)$.

14. (a) State and prove Lagrange's theorem.

Or

(b) State and prove the principle of cross classification.

15. (a) Determine the odd primes p for which 3 is a quadratic residue and those for which it is a nonresidue.

Or

(b) State anti prove the reciprocity law for Jacobi symbols.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

Each question carries 10 marks.

16. State and prove the five properties of the Euler totient function.

17. (a) Prove that $\sum_{n \le x} \frac{1}{n} = \log x + C + 0\left(\frac{1}{x}\right)$.

(b) Prove that the set of lattice points visible from the origin has density $6/\pi^2$.

18. Let p_n denote the nth prime. Prove that the following asymptotic relations are logically equivalent :

(a) $\lim_{x \to \infty} \frac{\pi(x) \log x}{x} = 1$

(b) $\lim_{x \to \infty} \frac{\pi(x) \log \pi(x)}{x} = 1$

(c) $\lim_{x \to \infty} \frac{p_n}{n \log n} = 1$.

19. Prove that the set of lattice points in the plane visible from the origin contains arbitrarily large square gaps.

20. State and prove Gauss lemma.

RW–924

Sub. Code
511513

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Mathematics

OBJECT ORIENTED PROGRAMMING AND C++

(CBCS 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give any two special features of C++ over C.

2. What is the output of the following program segment?

   n=7

   cout<<n++<<n<<n–;

3. Can objects appear as function arguments? Give an example.

4. What is inline function?

5. Write the uses of pointers in C++.

6. State the purpose of new operator.

7. What is function overloading?

8. What are the advantages of polymorphism?

9. What is virtual function?

10. How can you define a derived class?

Part B

$(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write the benefits of object oriented programming.

Or

(b) Describe the bitwise operators available in C++.

12. (a) How classes are declared in C++? Explain with an example.

Or

(b) What is a destructor? Write the rules for destructor.

13. (a) Write a program to count and print the frequency of occurrence of vowels in a given sentence.

Or

(b) List out the problems with pointer reference and explain.

14. (a) What is operator overloading? Why it is necessary to overload and operator?

Or

(b) Write a program to overload the + operator to provide string addition.

15. (a) Write about abstract base class.

Or

(b) Explain dynamic binding with an example.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. What are the data types supported by C++? Explain them with suitable examples.

17. Define a class to represent a bank account with data members – name of the depositor, account number, type of account and member functions – deposit amount and withdraw amount. Show name and balance. Write a program to test this class.

18. Explain copy constructor with an example program.

19. Write a program to perform complex number arithmetic operations using operator overloading.

20. Explain the different types of inheritance supported by C++ with examples.

RW-925

Sub. Code
511301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Mathematics

MECHANICS

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define non-holonomic constraints.

2. Define virtual displacement.

3. Write the standard form of Lagrange's equations for a holonomic system.

4. State any two properties of conservative system.

5. State the Brachistochrome problem.

6. Sate Hamilton's principle.

7. Write Jacobi's form of the principle of least action.

8. State Liouville's system.

9. Show that identity transformation is canonical.

10. State Lagrange's brackets.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b)

11. (a) Derive Lagrangian form of D'Alermbert's principle.

Or

(b) Discuss the kinetic energy of a system.

12. (a) Find the differential equations of motion for a spherical pendulum of length l?

Or

(b) Explain Integrals of the motion.

13. (a) Find the stationary values of the functions $f = z$ subject to the constraints.

$$\phi_1 = x^2 + y^2 + z^2 - 4 = 0$$

$$\phi_2 = xy - 1 = 0$$

Or

(b) Obtain modified Hamilton's principle.

14. (a) Explain pfaffian differential forms.

Or

(b) State and prove jacobi's theorem.

15. (a) Show that the transformation $\theta = \frac{1}{2}(q^2 + p^2), p = -\tan^{-1}\frac{q}{p}$ is cononical.

Or

(b) Write a short note t homogeneous canonical transformation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Explain generalized coordinates.

    (b) Explain holonomic constraints.

17. (a) Derive the largange's equation.

    (b) A double pendulum consists of two praticles suspended by massless rods. Assuming that all motion takes place in a vertical plane. Find the differential equations of motion and linearize these equations and assuming small motions.

18. Derive the Jacobi's form of the principle of least action.

19. State and prove 'stackel's theorem.

20. Write brief explain to poission's brackets.

RW–926

Sub. Code
511302

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Mathematics

TOPOLOGY

(CBCS 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Each questions carries 2 marks.

1. Define subspace topology.

2. State uniform limit theorem.

3. Define quotient topology.

4. Is local connectedness implies connectedness. Justify your answer.

5. Define compact space.

6. Define complete metric space.

7. Verify whether the second countable axiom is true in R^w with respect to uniform topology.

8. Let (x,d) be a metric space and $A \subset X$ show that $d(x,A) = 0$ iff $x \in \overline{A}$.

9. Define complete regular space.

10. Define stone- cech compactification at a space.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Let X be a space. Then prove that the following are equivalent

(i) $\phi \,\&\, X$ are closed.

(ii) Arbitrary intersections of closed sets are closed.

(iii) Finite union of closed sets are closed.

Or

(b) Let Y be a subspace of X. If $\cup$ is open in Y and Y is open in X, then prove that $\cup$ is open in X.

12. (a) State and prove sequence lemma.

Or

(b) Prove that the components of X are connected disjoint subsets of X whose union is X, such that each connected subset of X intersects only one of them.

13. (a) Prove that the image of a connected space under a continuous map is connected.

Or

(b) Let (x,d) be a metric space. Prove that there is an isometric imbedding of X into a complete metric space.

14. (a) Suppose X is countable basis then prove that every open covering of X contains a countable subcollection covering X.

Or

(b) Prove that every compact Hausdroff space is normal.

15. (a) Prove that a subspace of a competely regular space is completely regular.

Or

(b) Let $A \subset X$. Let $f : A \to Z$ be a continuous map of A into Hausdroff space Z. Prove that there is at most one extension of f to a continuous function $g : \overline{A} \to Z$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let X and Y be topological space, let $f : x \to y$. Then prove the following conditions are equivalent :

(a) f is continuous

(b) for every subset A of X one has $f(\overline{A}) e \overline{f(A)}$.

(c) For every closed set B in Y, the set $\bar{f}(B)$ is closed in X.

17. Prove that cartesian product of connected space is connected.

18. Let X be a metizable space prove that the following are equivalent

    (a) X is compact

    (b) X is limit point compact

    (c) X is sequentially compact

19. State and prove Urysohn's metrication theorem.

20. Let X be a completely regular, let $\beta(x)$ be its stone-cech compactification. Then prove every bounded continuously real-valued function on X can be unequally extended to a continuous real-valued function $\beta(x)$.

RW–927

Sub. Code
511303

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Mathematics

DIFFERENTIAL GEOMETRY

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define functions of class 'm'.

2. Prove that a necessary and sufficient condition that a curve to be a straight line is that $K = 0$ at all points.

3. Define osculating circle.

4. Define Helix.

5. Write the parametric equations of a surface.

6. Define singularity of the surface and types o singularity.

7. Define Geodesics.

8. State the necessary and sufficient condition that the curve $v = c$ be a geodesic.

9. Define Geodesic curvature.

10. Write Liouville's formula.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Calculate the curvature and torsion of the cubic curve $\bar{r} = (u, u^2, u^3)$

Or

(b) Prove that the length of the common perpendicular 'd', between the tangents at two near points distance 's' apart is approximately $d = \frac{K\tau s^2}{12}$.

12. (a) If the radius of spherical curvature is constant, prove that the curve either lies on a sphere or has constant curvature.

Or

(b) Show that the involutes of a circular helix are plane curves.

13. (a) Prove that the metric is invariant under the parametric transformation.

Or

(b) Prove that the metric is a positive quadratic form in dy, dv.

14. (a) Prove that the curves of the family $v^3/u^2 =$ constant are geodesics on S surface with metric $v^2 du^2 = 2uvdudv + 2u^2 dv^2, (u > 0, v > 0)$.

Or

(b) Prove that every Helix in a cylinder is a geodesic.

15. (a) Prove that on a paraboloid of revolution, every geodesic other than a meridian intersect, itself infinitely often.

Or

(b) Prove that the Gussian curvature of the surface given by $Z = f(x, y)$ is $(rt - s^2)(1 + p^2 + q^2)$, where $p = \partial z / \partial x,\ q = \dfrac{\partial z}{\partial y}\ r = \partial^2 z / dx^2, p = \dfrac{\partial^2 z}{\partial x\, \partial y},\ t = \partial^2 z / \partial y^2$

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Obtain a formula for the arc length of a path given interms of a parameter and also show that the length of one complete turn of the helix $\vec{r} = (a \cos u,\ a \sin u,\ bu),\ a > 0, -\infty < u \leq \infty$ is $2\pi c$, where $C = \sqrt{a^2 + b^2}$.

17. If the curve lies on the sphere then prove that ρ and τ are connected by the relation $\dfrac{d}{ds}(\tau\ p') + \dfrac{\rho}{\sigma} = 0$.

18. If surfaces $s \,\&\, s'$ are isometric then prove that there exists a correspondence $u' = \phi(u, v), v' = n\ u(u, v)$ between their parameters, where $Q \,\&\, \mu$ are single values and heave non-vanishing Jacobian such that the metric of S transforms into the metric of 's'. Find a surface of revolution which is isometric with a region of the right helicoid.

19. Let get for continuous for $0 < t < 1$ and if $\int_0^1 v(t)\, g(t)\, dt = 0$ for all admissible function then prove that $g(t) = 0$.

20. State and prove Gauss Bonnet theorem.

RW–928

Sub. Code
511304

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Mathematics

MULTIVARIATE CALCULUS

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define total derivative of a function at a point.

2. Define contraction mapping.

3. Define projection.

4. Give an example in which $D_{ij}f \neq D_{ji}f$ at some point, although both derivatives exist.

5. Define primitive map.

6. Define k – form in an open set.

7. Define oriented affine mapping.

8. Define affine k – chain.

9. Define closed k – form.

10. State the divergence theorem.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Suppose E is an open set in $\mathbb{R}^n$, $f : E \to \mathbb{R}^m$, and f is differentiable at a point $x \in E$. Then show that the partial derivatives $(D_j f_i)(x)$ exist, and

$$f'(x)e_j = \sum_{i=1}^{m} (D_j f_i)(x)\, u_i, \quad (1 \le j \le n).$$

Or

(b) If X is a complete metric space and if ϕ is a contraction of X into X. Prove that ϕ has a unique fixed point.

12. (a) If $[A]$ and $[B]$ are n by n matrices, then show that $\det([B][A]) = \det([A][B])$.

Or

(b) Suppose f is defined in an open set $E \subset \mathbb{R}^2$, prove that if $f \in \mathscr{C}''(E)$ then $D_{21} f = D_{12} f$.

13. (a) For every $f \in \mathscr{C}(I^k)$, show that $L(f) = L'(f)$.

Or

(b) Suppose $\omega = \sum_I b_I(X)\, dx_I$ is the standard presentation of a k – form ω in an open set $E \subset \mathbb{R}^n$. If $\omega = 0$ in E prove that $b_I(x) = 0$ for every increasing k–index I and for every $x \in E$.

14. (a) If w is a k-form of class $\mathscr{C}''$ in an open set $E \subset \mathbb{R}^n$, then prove that $d^2\omega = 0$.

Or

(b) If σ is an oriented rectilinear k–simplex in an open set $E \subset \mathbb{R}^n$ and if $\overline{\sigma} = \varepsilon\sigma$ then prove that $\int_{\overline{\sigma}} \omega = \varepsilon \int_{\sigma} \omega$ for every k–form ω in E.

15. (a) State and prove Green's theorem.

Or

(b) State and prove Stoke's formula.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove inverse function theorem.

17. State and prove Rank theorem.

18. Suppose K is a compact subset of $\mathbb{R}^n$ and $\{V_\alpha\}$ is an open cover of K. Prove that there exist functions $\psi_1, \cdots, \psi_s \in \mathscr{C}(\mathbb{R}^n)$ such that

(a) $0 \le \psi_1 \le 1$ for $i = 1, 2, \cdots, s$;

(b) each ψ_1 has its support in some V_α, and

(c) $\sum_{i=1}^{s} \psi_i(x) = 1$ for every $x \in K$. Prove also that if $f \in \mathscr{C}(\mathbb{R}^n)$ and the support of f lines in K then $f = \sum_{i=1}^{s} \psi_1 f$.

19. Suppose T is a $\mathscr{C}'$ – mapping of an open set $E \subset \mathbb{R}^n$ into an open set $V \subset \mathbb{R}^m$, Φ is a k–surface in E, and ω is a k–form in V. Prove that $\int_{T\Phi} \omega = \int_{\Phi} \omega_T$.

20. Prove that closed forms are exact in convex sets.

RW–929

Sub. Code
511515

M.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Mathematics

IMAGE PROCESSING AND PATTERN RECOGNITION

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What are the types of images?

2. List out the properties of 2D convolution.

3. State the purpose of edge detection.

4. What is the use of closing operations?

5. Define : Compression ratio.

6. Classify the image compression schemes.

7. What are the two categories of pattern recognition?

8. What is supervised learning?

9. What is syntactic pattern recognition?

10. What are the types of grammar?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is sampling and quantization?

Or

(b) Explain the need of histogram equilization.

12. (a) Describe the basic mathematical approaches used for edge detection.

Or

(b) Describe the opening operation and its properties.

13. (a) Write about contour coding.

Or

(b) Explain the usage of quadrees.

14. (a) Describe the basic concepts of pattern recognition.

Or

(b) Illustrate the generalized decision function.

15. (a) Write about multiprototypes.

Or

(b) How can you formulate the syntactic pattern recognition problem?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the gray level transformation with an example.

17. Describe the basic morphological operations.

18. Draw the standard image compression model and explain the functions of each block in the model.

19. What are decision functions? How can you implement it?

20. Describe the k-means algorithm and isodata algorithm.

———————